

Worksheet 3A: Quadratics — FOIL and Expanding

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

Part A: The FOIL Method — Basic Mechanics (12 problems)

FOIL stands for **F**irst, **O**uter, **I**nnner, **L**ast — a method for multiplying two binomials.

Example: $(x + 3)(x + 2)$

First: $x * x = x^2$

Outer: $x * 2 = 2x$

Inner: $3 * x = 3x$

Last: $3 * 2 = 6$

Combine: $x^2 + 2x + 3x + 6 = x^2 + 5x + 6$

Expand each expression completely. Show your F-O-I-L steps for the first few.

1. $(t + 2)(t + 5)$
2. $(p + 6)(p + 3)$
3. $(m + 4)(m + 7)$
4. $(r + 1)(r + 9)$
5. $(k + 8)(k + 2)$
6. $(n + 5)(n + 5)$ — *What do you notice about this one?*
7. $(w + 3)(w + 3)$
8. $(s + 10)(s + 1)$
9. $(h + 6)(h + 6)$
10. $(u + 4)(u + 4)$
11. What pattern do you see when both binomials are identical, like $(x + a)(x + a)$?
12. Complete this rule: $(x + a)^2 = x^2 + \underline{\hspace{2cm}} + a^2$

Part B: WAIT — Exponents Don't Distribute Over Addition! (8 problems)

Common Mistake Alert! Many students think $(a + b)^2 = a^2 + b^2$. **This is WRONG.** $(a + b)^2$ means $(a + b)(a + b)$, which you must FOIL out.

Let's prove this with actual numbers.

13. Let $a = 3$ and $b = 4$. - (a) Calculate $(a+b)^2 = (3+4)^2 = 7^2 =$ _____
 - (b) Calculate $a^2 + b^2 = 3^2 + 4^2 = 9 + 16 =$ _____ - (c) Are they equal? _____
14. Now let $a = 2$ and $b = 5$. - (a) $(a+b)^2 =$ _____ - (b) $a^2 + b^2 =$ _____
 - (c) What's the difference between the two answers? _____
15. Expand $(a+b)^2$ using FOIL: $(a+b)^2 = (a+b)(a+b) =$ _____
16. The correct expansion is $a^2 + 2ab + b^2$. What's the "missing piece" when people incorrectly write $a^2 + b^2$?
17. Using $a = 3$ and $b = 4$ again:
 (a) Calculate $2ab = 2(3)(4) =$ _____
 (b) Verify: $a^2 + 2ab + b^2 = 9 + 24 + 16 =$ _____
 (c) Does this match $(a+b)^2 = 49$? _____
18. Does the same mistake happen with subtraction? Test $(a-b)^2$ vs $a^2 - b^2$ with $a = 5$, $b = 2$:
 (a) $(a-b)^2 = (5-2)^2 =$ _____
 (b) $a^2 - b^2 = 25 - 4 =$ _____
 (c) Are they equal? _____
 (d) What is the correct expansion of $(a-b)^2$? _____
19. Here's another trap. Is $(a+b)^3 = a^3 + b^3$? Test with $a = 2$, $b = 1$:
 (a) $(a+b)^3 = (2+1)^3 = 3^3 =$ _____
 (b) $a^3 + b^3 = 8 + 1 =$ _____
 (c) Are they equal? _____
20. Complete the rule: **Exponents distribute over** _____
but NOT over _____. *(Hint: $(ab)^2 = a^2b^2$ works, but $(a+b)^2 \neq a^2 + b^2$)*

Part C: FOIL with Negative Numbers (12 problems)

When negatives appear, keep careful track of signs.

Example: $(x-4)(x+3)$

First: $x * x = x^2$

Outer: $x * 3 = 3x$

Inner: $-4 * x = -4x$

Last: $-4 * 3 = -12$

Combine: $x^2 + 3x - 4x - 12 = x^2 - x - 12$

Expand completely:

21. $(t - 3)(t + 5)$
22. $(p + 4)(p - 2)$
23. $(m - 6)(m + 1)$
24. $(r - 2)(r - 5)$
25. $(k - 7)(k - 3)$
26. $(n - 4)(n - 4)$ — *Squaring a binomial with subtraction*
27. $(w - 8)(w - 8)$
28. $(s + 5)(s - 5)$ — *What happens to the middle term?*
29. $(h + 9)(h - 9)$
30. $(u + 7)(u - 7)$
31. What pattern do you see when the binomials are $(x + a)(x - a)$?
32. Complete this rule: $(x + a)(x - a) = \underline{\hspace{2cm}}$

Part D: FOIL with Coefficients (10 problems)

When the variable has a coefficient, apply FOIL the same way.

Example: $(2x + 3)(x + 4)$

First: $2x * x = 2x^2$

Outer: $2x * 4 = 8x$

Inner: $3 * x = 3x$

Last: $3 * 4 = 12$

Combine: $2x^2 + 8x + 3x + 12 = 2x^2 + 11x + 12$

Expand completely:

33. $(2t + 1)(t + 3)$
34. $(3p + 2)(p + 4)$
35. $(2m - 3)(m + 5)$
36. $(4r + 1)(r - 2)$
37. $(3k - 2)(2k + 1)$
38. $(5n + 2)(2n - 3)$
39. $(2w + 3)(2w + 3)$ — *Squaring with coefficients*
40. $(3s - 1)(3s - 1)$
41. $(4h + 5)(4h - 5)$ — *Difference of squares with coefficients*
42. $(2u - 7)(2u + 7)$

Part E: Patterns Worth Memorizing (8 problems)

These patterns appear so often that recognizing them instantly saves time.

Pattern 1: Perfect Square Trinomials

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

The middle term is always **twice** the product of the two terms.

Pattern 2: Difference of Squares

$$(a + b)(a - b) = a^2 - b^2$$

The middle terms cancel out, leaving only the difference of squares.

Use the patterns to expand quickly:

- 43. $(t + 8)^2$
- 44. $(p - 5)^2$
- 45. $(m + 11)(m - 11)$
- 46. $(3r + 4)^2$
- 47. $(5k - 2)^2$
- 48. $(6n + 1)(6n - 1)$
- 49. $(2w + 9)^2$
- 50. $(4s - 3)(4s + 3)$

Part F: Discovering Pascal's Triangle (12 problems)

Let's expand powers of $(a + b)$ and look for patterns in the coefficients.

- 51. $(a + b)^0 = 1$ Coefficients: **1**
- 52. $(a + b)^1 = a + b$ Coefficients: **1, 1**
- 53. Expand $(a + b)^2$ using FOIL. $(a + b)^2 =$ _____ Coefficients of a^2 , ab , and b^2 : _____, _____, _____
- 54. Expand $(a + b)^3$ by computing $(a + b) \cdot (a + b)^2$. *Use your answer from problem 53 and distribute $(a + b)$ through it. * $(a + b)^3 =$ _____ Coefficients: _____, _____, _____, _____
- 55. Expand $(a + b)^4$ by computing $(a + b) \cdot (a + b)^3$. $(a + b)^4 =$ _____ Coefficients: _____, _____, _____, _____, _____

56. Arrange your coefficients in a triangle:

					$(a + b)^0$
		1	1		$(a + b)^1$
	1	2	1		$(a + b)^2$
1	3	3	1		$(a + b)^3$
1	?	?	?	1	$(a + b)^4$

Fill in the missing numbers for row 4.

57. Look at how each number relates to the two numbers above it. Describe the pattern:

58. Using the pattern, predict row 5 (the coefficients for $(a + b)^5$): _____,
_____, _____, _____, _____

59. This triangle is called _____'s **Triangle**. (*Named after a 17th-century French mathematician, though mathematicians in Persia, China, and India discovered it centuries earlier.*)

60. Expand $(a+b)^5$ using your predicted coefficients: $(a+b)^5 =$ _____

61. **Bonus connection:** The number in row n , position k tells you "how many ways can you choose k items from n items?" Row 4, position 2 has the number **6**. Verify: List all ways to choose 2 letters from A, B, C, D: _____
(*There should be exactly 6 ways!*)

62. Using the triangle, how many ways can you choose 3 items from 5 items? (*Look at row 5, position 3*)

Part G: Beyond Two Terms (8 problems)

For multiplying larger expressions, we distribute systematically.

Example: $(x + 2)(x^2 + 3x + 1)$

Distribute x : $x \cdot x^2 + x \cdot 3x + x \cdot 1 = x^3 + 3x^2 + x$

Distribute 2: $2 \cdot x^2 + 2 \cdot 3x + 2 \cdot 1 = 2x^2 + 6x + 2$

Combine: $x^3 + 3x^2 + 2x^2 + x + 6x + 2 = x^3 + 5x^2 + 7x + 2$

Expand completely:

63. $(t + 3)(t^2 + 2t + 4)$

64. $(p - 2)(p^2 + 4p + 1)$

65. $(m + 1)(m^2 - m + 1)$ — *Something interesting happens here!*

66. $(r - 1)(r^2 + r + 1)$ — *And here!*

67. $(2k + 1)(k^2 - 3k + 2)$

68. $(n + 2)(n + 3)(n + 1)$ — *Do two at a time*

69. Expand $(w - 1)^3$ by first finding $(w - 1)^2$, then multiplying by $(w - 1)$.

70. Expand $(2t + 1)^3$ using the same method.